
DIPTIMAN BORA

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EDUCATION

The University of Arizona

Bachelor of Science in Physics (*with Honors*) & Data Science (*Computing Emphasis*), Minor in Computer Science

Honors: Global Wildcat Award (\$23,000 per year), US-ATLAS ATC Award

Member, W.A. Franke Honors College

Tucson, AZ

Expected May 2028

GPA: 3.946/4.00

RESEARCH EXPERIENCE

Dr. Nikolay Golubev, Department of Physics, The University of Arizona

October 2025 - present

- Applying theoretical frameworks in molecular and ultrafast dynamics, including Born–Oppenheimer approximation and the exact factorization of the molecular wavefunction, toward future modeling work

Dr. Kenneth Johns, Department of Physics, The University of Arizona

October 2024 - present

Project Reports: [Google Drive](#)

- Developed more than 10 Python scripts to test FPGA firmware being developed for ATLAS Liquid Argon calorimeter readout at CERN LHC, generating test data for longer-duration testing of these boards via SSH
- Applied ML models (MLPs, Graph Neural Networks) to differentiate long-lived particles (LLPs) in ATLAS Muon Spectrometer Run 2 data, optimizing GPU-based training on Monte Carlo and real datasets, and now preparing for new Run 3 data

GRANTS & ACHIEVEMENTS

US-ATLAS ATC Award (Selected for Funding)

2026

Awarded funding to support research with the ATLAS experiment at Brookhaven National Laboratory.

Dean's List with Distinction

Spring 2025, Fall 2025

Recognition for outstanding academic achievement by term for having GPA of 4.0.

Dean's List

Fall 2024

Recognition for outstanding academic achievement by term for having GPA 3.500-3.999 (3.76).

Global Wildcat Award

Fall 2024 - present

A merit-based scholarship for meritorious incoming undergraduate students. Awarded for four academic years by The University of Arizona.

ACADEMIC EXPERIENCE

PHYS332 Final Project: Motion of a Charged Particle in a Dipole Field

January 2026 - present

Personal GitHub repository: github.com/D-man165

Instructor: Dr. Fulvio Melia

- Created an iterating code to find the trajectory of charged particle in a dipole field at Origin

ISTA322 Final Project: Global Food Insecurity

August 2025 - December 2025

Personal GitHub repository: github.com/D-man165

Instructor: Dr. Sara Riazi

- Developed an ETL pipeline that integrates multiple real-world data sources to analyze how moderate and severe food insecurity varies across countries, and how it relates to income and population density in multiple years
- Acquired data via API requests and csv files over 3000 lines long, cleaned and normalized them into an SQL database, and then analyzed relationships between insecurity and population and income as a personal web app using Streamlit

TEACHING EXPERIENCE

Preceptor: PHYS 241: Introductory Electricity and Magnetism

Department of Physics, The University of Arizona

Fall 2025

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- Oversaw more than 250 students during their group work for PHYS241 (E&M) class, along with 4 other preceptors and TAs to resolve doubts and issues students might face which requires having sufficient knowledge of the subject and asking thoughtful questions

Preceptor: PHYS 111: Introductory Studio Physics 2

Department of Physics, The University of Arizona

January 2026 - present

- I work alongside TAs in weekly lab sessions to resolve student questions, troubleshoot experimental issues, and demonstrate physics experiments, requiring clear explanatory skills

PROJECTS

CERN GitLab, Personal GitHub

CERN gitlab repository: gitlab.cern.ch/dbora

Personal GitHub repository: github.com/D-man165

- Documented and developed various Machine Learning algorithms used to identify LLPs such as MLPs, Graph Convolution Networks and Graph Attention Networks with attached plots
- Created projects such as MLP neural network, Linear Regression from scratch as an introduction to ML

TECHNICAL SKILLS

Programming Skills:	Machine Learning, Data Analysis and Visualization, Object Oriented Programming, Data Pipeline Implementation
Programming Languages:	Python, SQL, Java
Operating Systems:	Linux, Windows
Software:	Git, Jupyter Notebook, L ^A T _E X, Command Line Interfaces
Hardware & Electronics:	Experience with Raspberry Pi, Arduino, BBC micro:bit and general electronic circuits

CERTIFICATIONS & TRAINING

[Stanford University & DeepLearning.AI Machine Learning Specialization](#)

Online course: Coursera, May 2025

[Data Analysis with Python by IBM](#)

Online course: IBM, March 2025

3D Printing Certification – University of Arizona CATalyst Studios

September 2024

[CS50's Introduction to Programming with Python](#)

Online course: HarvardX, Dec. 2023

ADDITIONAL SKILLS

Transferable skills:	Critical Thinking and Curiosity, Analytical Skills, Teaching
Languages:	English (Professional Speaking Capacity); Assamese (Native); Hindi (Native)